

ROMA, Australia

WMO: 945150

Lat: 26.544S

Lon: 148.778E

Elev: 1010

StdP: 14.17

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.5	33.0	16.3	13.0	64.1	19.5	15.2	61.1	20.1	63.4	18.5	63.5	2.6	0	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	100.8	67.3	97.9	67.3	95.3	67.1	74.7	86.0	73.4	84.5	72.2	83.6	11.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	123.7	77.9	70.5	116.7	77.1	69.1	110.9	76.8	39.0	85.8	37.7	84.2	36.7	83.4	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.8	16.1	25.8	106.1	1.7	3.6	24.6	108.7	23.7	110.8	22.8	112.9	21.6	115.5		
(5)			26.1	79.6	3.2	5.9	23.7	83.8	21.9	87.3	20.0	90.6	17.7	94.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	81.6	79.2	76.0	68.3	59.9	54.7	53.2	56.3	64.4	71.0	76.1	79.6		
	DBStd	11.52	4.87	5.23	4.75	5.32	5.80	5.86	5.64	6.31	6.93	6.38	5.99	5.51		
	HDD50	61	0	0	0	0	2	20	28	11	0	0	0	0		
	HDD65	1282	0	0	1	29	177	310	367	281	95	20	2	0		
	CDD50	6743	978	817	807	548	309	162	128	206	433	652	784	919		
	CDD65	2487	513	397	343	127	18	2	2	11	77	207	336	454		
	CDH74	28221	6082	4342	3456	1325	257	25	40	240	1147	2451	3667	5189		
	CDH80	13445	3258	2178	1523	363	32	2	1	56	410	1074	1808	2740		
(14) Wind	WSAvg	7.6	9.1	8.5	7.7	6.4	5.7	5.9	6.2	7.0	7.9	8.8	9.3	9.3		
(15) Precipitation	PrecAvg	24.3	3.2	2.4	2.3	1.5	1.8	1.1	1.5	1.3	1.1	2.0	2.5	3.1		
	PrecMax	35.3	8.5	9.9	10.2	8.2	9.4	4.3	7.1	4.1	3.7	8.3	5.9	10.6		
	PrecMin	15.1	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3		
	PrecStd	5.7	2.1	2.0	2.6	2.1	1.8	1.2	1.7	0.9	1.1	1.6	1.9	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.5	103.3	98.6	90.3	83.7	76.7	78.4	85.8	94.4	98.1	102.1	104.6		
		MCWB	69.4	68.4	66.7	64.6	61.0	59.4	57.5	59.3	62.5	64.8	66.3	66.5		
	2%	DB	99.4	98.4	94.3	86.6	79.7	73.7	74.2	79.9	88.6	93.3	97.6	100.2		
		MCWB	68.4	67.7	67.5	63.9	60.8	58.5	55.9	57.3	60.2	63.6	65.8	66.9		
	5%	DB	96.6	94.7	90.7	84.2	76.7	71.3	70.9	75.9	84.6	89.8	93.8	96.2		
		MCWB	68.3	68.5	66.8	62.8	59.5	57.5	54.8	55.4	59.4	62.8	64.8	66.8		
	10%	DB	93.7	91.3	87.9	81.7	74.0	68.3	67.8	72.3	80.8	86.1	89.9	92.8		
		MCWB	68.2	68.4	66.2	62.2	58.2	56.1	53.5	53.9	58.2	61.9	64.3	66.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.5	76.3	74.0	71.0	66.9	65.5	63.4	63.6	67.7	75.2	74.1	74.4		
		MCDB	89.3	86.6	83.8	79.3	74.8	70.9	71.9	78.4	80.9	85.9	85.7	87.8		
	2%	WB	74.6	74.5	72.3	68.3	64.4	61.9	59.5	60.3	64.5	68.4	70.7	72.7		
		MCDB	86.1	85.4	82.2	79.2	72.3	68.7	67.6	72.6	77.3	81.3	82.9	85.9		
	5%	WB	73.2	73.3	71.0	66.7	62.4	59.9	57.4	58.1	62.5	66.5	69.1	71.4		
		MCDB	84.9	83.5	81.3	77.3	71.8	67.2	65.7	70.0	77.2	78.5	81.7	84.0		
	10%	WB	71.9	72.1	69.7	64.9	60.6	58.0	55.5	56.1	60.9	64.8	67.7	70.2		
		MCDB	84.1	82.6	80.4	76.1	70.7	65.3	64.9	68.5	75.1	78.1	81.4	82.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	21.4	23.2	26.0	27.4	25.8	28.7	30.9	29.5	27.0	23.8	22.9		
		MCDBR	25.8	26.5	26.8	28.3	29.1	27.7	31.5	34.2	33.3	31.5	29.3	28.4		
		MCWBR	7.8	8.1	8.4	11.4	13.5	14.5	16.2	16.0	13.6	12.0	10.0	8.9		
		MCDBR	20.3	19.7	20.0	21.6	22.2	21.5	23.6	26.3	25.6	23.8	21.8	21.1		
	5% WB	MCWBR	7.5	6.7	7.2	9.3	10.8	12.2	13.6	13.1	11.8	10.9	8.3	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.365	0.355	0.334	0.318	0.298	0.290	0.283	0.295	0.331	0.351	0.370	0.369		
	taud		2.518	2.555	2.606	2.607	2.605	2.640	2.636	2.585	2.470	2.449	2.447	2.487		
	Ebn at Noon		310	308	306	296	289	285	292	300	301	306	306	310		
	Edh at Noon		36	34	30	28	25	23	24	28	34	37	38	37		
(44) All-Sky Solar Radiation	RadAvg		2304	2143	1901	1615	1303	1092	1235	1551	1882	2138	2272	2303		
	RadStd		167	174	112	95	59	86	89	98	130	168	222	204		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) +2.30	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+2.30	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page